CHAPTER 7

HIGHER-ORDER MUTUAL INFORMATION (HIGHER-MI)

FORMULATION, KSG ESTIMATION, AND BENCHMARK EVALUATION

7.1. INTRODUCTION AND MOTIVATION

In the theoretical analysis of deep neural networks, the term **grokking** describes the delayed and abrupt emergence of generalisation long after the model has achieved perfect training accuracy. Under standard training schedules with strong weight decay, the training error vanishes within a few hundred epochs, while the test accuracy remains close to random guessing for thousands of epochs before sharply ascending towards 100%.

Because grokking occurs silently without early improvements in test loss, identifying internal network representations that can forecast this transition is of fundamental importance. In this chapter, we formulate, implement, and rigorously evaluate **Predictor 7** of our unified benchmark, designated as the **Higher-Order Mutual Information (Higher-MI)** predictor.

This predictor adapts the higher-order interaction framework proposed by Pomarico et al. (2025) using **O-information** (denoted by Ω). Whereas traditional Shannon mutual information quantifies pairwise dependencies between two random variables, O-information determines whether a triplet of continuous variables is dominated by *redundancy* (shared overlapping information) or *synergy* (information that arises only when the variables are considered jointly). Our primary empirical objective is to determine whether the population O-information curve $\Omegapop(t)$ computed across held-out test examples exhibits a global maximum strictly *before* the grokking epoch G, thereby providing a reliable early-warning indicator.

7.2. MODULAR ARITHMETIC TASK AND THE GROK EPOCH

7.2.1. Task Formulation and Dataset Partitioning

Let the network be trained to perform modular addition of two integers modulo a prime $p = 113$. Given two input operands $a, b \in \{0, 1, \ldots, 112\}$, the unique correct output token c is determined by the congruence relation:

$$\mathbf{c \equiv (a + b) \mod 113, c \in \{0, 1, \ldots, 112\}} \quad ...(1)$$

For example, $(23 + 45) \bmod 113 = 68$, and $(100 + 20) \bmod 113 = 120 - 113 = 7$. The total combinatorial space consists of $113 \times 113 = 12{,}769$ input pairs. As standard in grokking studies, the dataset is partitioned into a 30% training set (3,830 pairs) used for gradient updates, and a 70% held-out test set (8,939 pairs) reserved strictly for evaluating generalisation.

7.2.2. Model Architecture and Checkpoint Schedule

The model is a 1-layer, 4-head transformer ($d_{\mathrm{model}} = 128$, head width 32, MLP width 512 with ReLU activation, learned positional embeddings, without LayerNorm). Training is performed full-batch for 40,000 epochs using AdamW (learning rate $\eta = 10^{-3}$, weight decay $\lambda = 1.0$, betas $\beta = (0.9, 0.98)$). Model states are saved every 100 epochs, yielding exactly 401 checkpoints per run across five independent random seeds (numbered 0 to 4).

7.2.3. The Grok Epoch (G)

The grok epoch G is defined as the earliest checkpoint epoch at which test accuracy exceeds 90%:

$$\mathbf{G = \min \{ t : Acc_{test}(t) > 0.90 \}} \quad ...(2)$$

The measured grok epochs across the five seeds are: **Seed 0: 12,987; Seed 1: 6,921; Seed 2: 9,510; Seed 3: 11,019; Seed 4: 23,747**. Because G varies substantially, an effective early warning must fire prior to G in each run and must fire later in runs that grok later.

7.3. REDUCTION OF LOGITS TO THREE SUMMARY VARIABLES

For each input pair, the network generates an unnormalised real score vector $\boldsymbol{\ell} \in \mathbb{R}^{113}$, where $\ell[j]$ denotes the **logit** assigned to answer class j. Note that logits are not probabilities; they are unconstrained real numbers that do not sum to unity. Softmax is not applied.

Because our task involves 113 valid answer classes, estimating continuous mutual information over a 113-dimensional distribution is intractable. To apply three-variable O-information, the 113 logits are reduced to three physically meaningful scalar summaries for each test example:

DEFINITIONS OF THE THREE LOGIT SUMMARIES:
- $\mathbf{F_0 = \blacksquare[c]}$: Logit assigned to the true correct answer class *c*.
- $\mathbf{F_1 = \max_{j \neq c} \blacksquare[j]}$: Highest logit among all incorrect classes (the runner-up).
- $\mathbf{F_2 = \&frac1;\{111\} \sum_{j \notin \{c, r\}} \blacksquare[j]}$: Arithmetic mean of the remaining 111 background wrong logits.

The classification is correct precisely when $F_0 > F_1$. The quantity $\mathbf{m = F_0 - F_1}$ is called the **classification margin**. A positive margin ($m > 0$) signifies a correct prediction, whereas $m < 0$ indicates an error.

Example 7.1. For the test pair (23, 45) under Seed 0, the correct answer is c = 68. At epoch 12,900, the network produces a margin m = +5.4, with class 54 as runner-up. State whether the prediction is correct and discuss its evolution.

Solution.
We are given that: $c = 68$, and the reported margin is $m = F_0 - F_1 = +5.4$.
By definition, the prediction is correct if and only if $F_0 > F_1$, which is equivalent to $m > 0$.
Since $m = +5.4 > 0$, we have $F_0 = F_1 + 5.4$. Thus, the score of the correct class 68 exceeds the highest competing wrong answer by 5.4 units. Therefore, the model's prediction is **correct**.
Physical Note: During training, this margin evolves from $m \approx -36.4$ at epoch 5,000 (wrong class 58 leading) to $m = +5.4$ at epoch 12,900, and reaches $m \approx +15.8$ at epoch 20,000. **Ans.**

7.4. MUTUAL INFORMATION AND O-INFORMATION

Mutual information $I(X; Y)$ quantifies the statistical dependence between continuous random variables X and Y. Measured in natural units (**nats**), $I(X; Y) = 0$ if and only if X and Y are independent. Conditional mutual information $I(X; Y \mid Z)$ measures the dependence remaining between X and Y when Z is already known.

For the three logit summaries (F_0, F_1, F_2), the **O-information** Ω is defined as the difference between pairwise and conditional mutual information:

$$\mathbf{\Omega(F_0, F_1, F_2) = I(F_0; F_1) - I(F_0; F_1 \mid F_2)} \quad ...(3)$$
- **Redundancy Dominated ($\Omega > 0$):** Pairwise mutual information exceeds conditional mutual information.
- **Synergy Dominated ($\Omega < 0$):** Conditional mutual information exceeds pairwise mutual information.

Note 1 (Distinction from Accuracy and Correlation). The sign of Ω describes the joint statistical structure of the three scores across examples; it is not a direct proxy for test accuracy or linear correlation. For example, at epoch 20,000 in Seed 0, F_0 and F_1 have a linear Pearson correlation of $r = 0.998$ across examples, while the population O-information is $\Omegapop \approx 0.36\text{ nats}$.

7.5. ESTIMATION BY KSG (NEAREST-NEIGHBOUR) METHOD

Because the joint probability density function $p(F_0, F_1, F_2)$ is unknown and fixed histogram binning is sensitive to bin widths, we adopt the non-parametric Kraskov–Stögbauer–Grassberger (KSG) estimator with $k = 4$ nearest neighbours under the Chebyshev (ℓ_∞ maximum) norm.

WORKING RULE 7.1: KSG ESTIMATION OF O-INFORMATION

Step 1. For each sample point $i \in \{1, \ldots, T\}$ in 3D space (F_0, F_1, F_2), find the Chebyshev distance ε_i to its 4th-nearest neighbour:

$\varepsilon_i = \max \{ |F_{0,i} - F_{0,j}|, |F_{1,i} - F_{1,j}|, |F_{2,i} - F_{2,j}| \}$.

Step 2. Around point i, count the number of other sample points lying *strictly within* radius ε_i in 1D subspaces $(N_{x,i}, N_{y,i}, N_{z,i})$ and 2D subspaces $(N_{xz,i}, N_{yz,i})$:

$N_{x,i} = \sum_{j \neq i} I(|x_i - x_j| < \varepsilon_i)$.

Step 3. Evaluate mutual information and conditional mutual information via the digamma function $\psi(u)$:

$I(X; Y) = \psi(k) + \psi(T) - \langle \psi(N_x + 1) + \psi(N_y + 1) \rangle$,

$I(X; Y \mid Z) = \psi(k) + \langle \psi(N_z + 1) - \psi(N_{xz} + 1) - \psi(N_{yz} + 1) \rangle$.

Step 4. Evaluate Ω across all 3! = 6 permutations of (F_0, F_1, F_2) and compute the arithmetic average.

Note 2 (The Strict Inequality Requirement). In the KSG formulation, the point setting the 4th-neighbour distance lies exactly on the boundary and must *not* be counted inside the marginal balls. Standard KD-tree queries default to inclusive checking ($\leq \varepsilon_i$), which erroneously inflates marginal counts. Querying with radius r = np.nextafter(eps, 0) enforces strict counting, restoring exact agreement with brute-force baselines.

7.6. POPULATION O-INFORMATION CURVE AND PEAK EVENT DETECTION

At each checkpoint t, the full held-out test set of $T = 8{,}939$ examples yields $8{,}939$ triplets (F_0, F_1, F_2). Computing the KSG estimate yields the population curve $\Omegapop(t)$ across all 401 checkpoints. The primary predictor event P is designated as the checkpoint corresponding to the global maximum:

$$P = \arg\max_t \ \&Omegapop;(t) \quad ...(4)$$

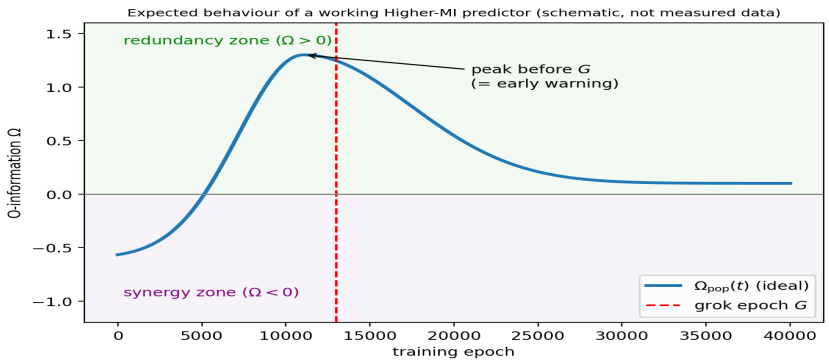


Figure 7.1: *Theoretical ideal schematic: O-information starting negative in the synergy zone, transitioning smoothly across zero, and peaking strictly before grok epoch G.*

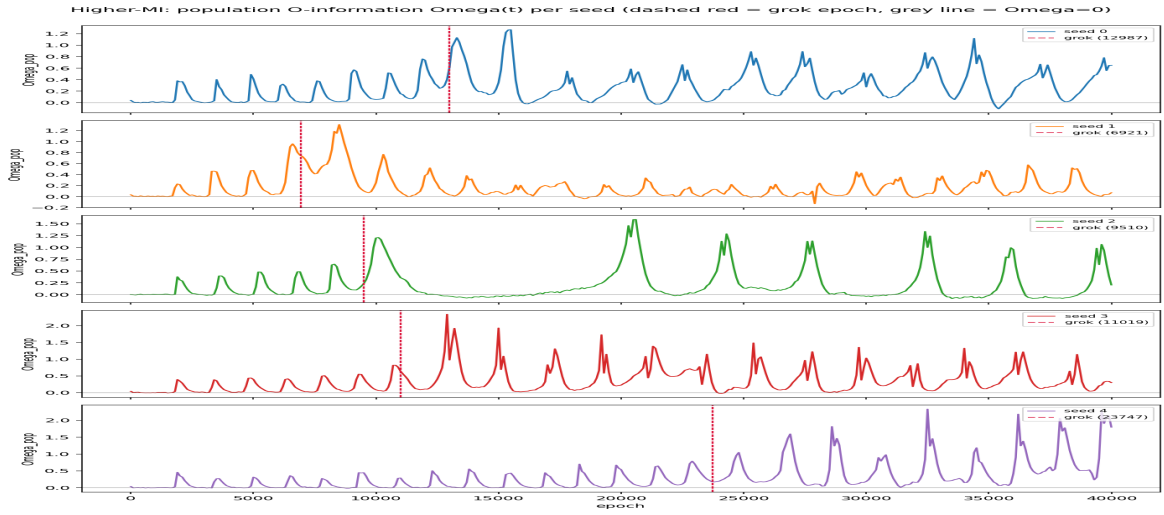


Figure 7.2: *Measured empirical population O-information $\Omegapop(t)$ across Seeds 0 to 4. In every seed, the curve oscillates repeatedly, exhibits a common spike at epoch 1,900, and reaches its global maximum after G (red dashed line).*

7.7. BENCHMARK SCORING CONDITIONS AND TIMING ANALYSIS

To ensure unbiased evaluation, the benchmark enforces three frozen statistical conditions:

WORKING RULE 7.2: FROZEN BENCHMARK SCORING CRITERIA
Condition 1 (Early-Warning Fraction): At least 80% of seeds (≥ 4/5) must satisfy $P_s < G_s - 100$.
Condition 2 (Spearman Rank Correlation): Monotonic rank correlation ρ between $\{G_s\}$ and $\{P_s\}$ must satisfy ρ ≥ 0.90.
Condition 3 (Mean Relative Lead): The mean relative lead across seeds must lie in [0.05, 0.90]:
mean_rel_lead = &frac1;{n} $\sum_{s=1}^{n} (G_s - P_s) / G_s \in [0.05, 0.90]$.

Seed	Grok Epoch (G)	Peak Epoch (P)	Timing Lag (P - G)	Ratio (P/G)	Relative Lead (G - P)/G
0	12,987	15,500	+2,513	1.19	-0.1935
1	6,921	8,500	+1,579	1.23	-0.2281
2	9,510	20,600	+11,090	2.17	-1.1661
3	11,019	12,900	+1,881	1.17	-0.1707
4	23,747	32,500	+8,753	1.37	-0.3686

Table 7.1: *Empirical Grok Epochs, Higher-MI Peak Events, and Timing Metrics Across Seeds.*

Example 7.2. For Seed 0, the model groks at epoch G = 12,987, while the population O-information reaches its global maximum at epoch P = 15,500. Compute the timing lag, the ratio P/G, and the relative lead, and state whether this run provides an early warning.

Solution.
We are given: $G = 12{,}987$, $P = 15{,}500$.
(i) **Timing Lag:** $P - G = 15{,}500 - 12{,}987 = +2{,}513\text{ epochs}$. The event is $2{,}513$ epochs late.
(ii) **Timing Ratio:** $P / G = 15{,}500 / 12{,}987 \approx 1.1935$.
(iii) **Relative Lead:** $\frac{G - P}{G} = \frac{12{,}987 - 15{,}500}{12{,}987} = \frac{-2{,}513}{12{,}987} \approx -0.1935$.
Since $P > G - 100$ (the event occurs after grokking), this seed fails the early-warning condition. **Ans.**

Example 7.3. Construct the rank difference table for G and P across the five seeds and calculate Spearman's rank correlation coefficient ρ.

Solution.
Let us assign ranks in ascending order (Rank 1 = smallest value):
• Seeds: (0, 1, 2, 3, 4)
• Grok epochs G: (12987, 6921, 9510, 11019, 23747) ⇒ Ranks R_G: (4, 1, 2, 3, 5)
• Peak epochs P: (15500, 8500, 20600, 12900, 32500) ⇒ Ranks R_P: (3, 1, 4, 2, 5)
• Differences $d_s = R_G - R_P$: $(4-3, 1-1, 2-4, 3-2, 5-5) = (+1, 0, -2, +1, 0)$
• Squared differences d_s^2: $(1, 0, 4, 1, 0) \implies \sum d_s^2 = 6$.
Using Spearman's formula for $n = 5$:
$\rho = 1 - [6 \sum d_s^2] / [n(n^2 - 1)] = 1 - [6 \times 6] / [5(25 - 1)] = 1 - 36 / 120 = 1 - 0.30 =$ **0.70**.
Since the required threshold is ρ ≥ 0.90, Condition 2 fails. **Ans.**

Benchmark Criterion	Required Threshold	Observed Metric	Official Status
Early-Warning Fraction	≥ 0.80 (≥ 4/5)	0/5 = 0.00	**FAIL**
Spearman Correlation (ρ)	≥ 0.90	0.70 (r = 0.898)	**FAIL**
Mean Relative Lead	[0.05, 0.90]	-0.4254	**FAIL**

Table 7.2: *Official Benchmark Evaluation Outcomes for Predictor 7 (Higher-MI).*

Final Verdict: All three criteria fail. Predictor 7 is formally declared **CLOSED, NEGATIVE**.

7.8. TEMPORAL SLIDING-WINDOW CONTROL AND TRANSFER ENTROPY

To test whether the failure of population O-information was an artifact of cross-sectional sampling across test inputs, two temporal controls were evaluated:

1. Sliding-Window Temporal O-Information ($\Omegatemp$): Computes O-information over mean logit trajectories in windows of length $W = 100$ checkpoints ($9{,}900$ epochs) with stride 10. In Seed 0, $\Omegatemp(t)$ begins near 1.55 nats at epoch 5,000, dips to 0.52 nats near epoch 10,000, and levels off between 1.20 and 1.45 nats. It is strictly positive across all runs, and three of five seeds tie at epoch 5,000, yielding $\rho = 0.00$.

2. Transfer Entropy: Evaluated across time lags $\tau \in \{1, \ldots, 10\}$. At lag 1, $\mathrm{TE}_{F_0 \to F_1}$ exceeds $\mathrm{TE}_{F_1 \to F_0}$ across all seeds, reflecting the shared upward drift of logits during late training rather than a predictive early warning.

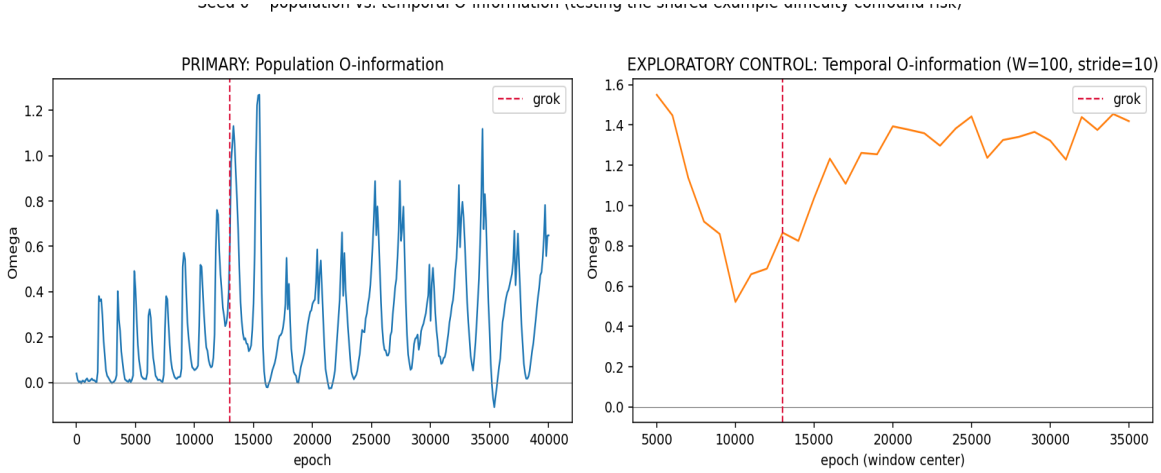


Figure 7.3: *Seed 0 comparison: Primary Population O-information $\Omegapop(t)$ (left) vs. Exploratory Sliding-Window Temporal O-information $\Omegatemp(t)$ (right).*

7.9. COMPARISON WITH THE SOURCE PAPER (POMARICO ET AL., 2025)

This negative benchmark verdict does not refute Pomarico et al. (2025). The differences are fundamental: (i) *Architecture:* Pomarico et al. evaluated a Matrix Product State (MPS) tensor network, whereas our benchmark employs a 1-layer 4-head transformer. (ii) *Dimensionality:* Their work studied a 3-class problem where logits mapped directly to (F_0, F_1, F_2), whereas modular arithmetic has 113 classes, requiring projection. (iii) *Objective:* The original authors demonstrated that O-information tracks representations around grokking; they did not claim that the global maximum serves as a strictly causal early warning under frozen correlation criteria.

7.10. CONSOLIDATED BENCHMARK STATUS (7 OF 8 PREDICTORS CLOSED)

With Predictor 7 closed, seven of the eight predictors in our evaluation order are now formally closed, all with negative verdicts (Table 7.3).

#	Predictor	Source Paper	Primary Event Rule & Key Metric	Status
1	L2 Norm	Nanda et al. (2023)	MA crossover fires at epoch ~125 (5/5 lead), ρ = 0.70	CLOSED (Neg)
2	Dropout Variance	Salah & Yevick (2025)	Variance peak arrives late (0/5 lead), ρ = 0.40	CLOSED (Neg)
3	Spectral Alignment	Canatar et al. (2021)	Alignment max arrives post-grok (0/5 lead), ρ = -0.30	CLOSED (Neg)
4	AGE (NC1 Collapse)	Papyan et al. (2020)	NC1 minimum arrives post-grok (0/5 lead), ρ = -0.40	CLOSED (Neg)
5	HTSR Alpha	Martin & Mahoney (2021)	Mean α min arrives late (0/5 lead), ρ = 0.90	CLOSED (Neg)
6	Weight-PCA	Yunis et al. (2024)	Effective rank min arrives post-grok (0/5 lead), ρ = 0.80	CLOSED (Neg)
7	**Higher-MI**	**Pomarico et al. (2025)**	**Pop. Ω max arrives post-grok (0/5 lead), ρ = 0.70**	**CLOSED (Neg)**

8	Commutator Defect	Pending	Final predictor in benchmark evaluation order	Next (Open)

Table 7.3: *Consolidated Status Across All Eight Grokking Predictors in the Unified Benchmark.*

EXERCISE 7.1 (QUESTIONS FOR REVIEW)

I. Theoretical and Conceptual Questions

1. Define O-information Ω for three continuous random variables. Explain the physical significance of positive ($\Omega > 0$) and negative ($\Omega < 0$) values.

2. Explain why the 113 output logits of the modular arithmetic transformer must be reduced to three summary variables (F_0, F_1, F_2), and give the physical meaning of each summary.

3. State the condition under which the KSG nearest-neighbour estimator requires strict inequality (<) rather than inclusive inequality (≤) during marginal point counting.

4. Explain the key methodological differences between the population O-information &Omegapop;(t) and the sliding-window temporal control &Omegatemp;(t).

5. Discuss why the empirical failure of Higher-MI in this benchmark does not contradict the conclusions of Pomarico et al. (2025).

II. Numerical and Analytical Problems

6. In Seed 3, the grok epoch is G = 11,019 and the population O-information peak is reached at epoch P = 12,900. Calculate: (i) the timing lag P − G, (ii) the timing ratio P/G, and (iii) the relative lead (G − P)/G.
[Ans. (i) +1,881 epochs; (ii) 1.1707; (iii) –0.1707]

7. In a synthetic 3D test set of T = 500 points, the Chebyshev distance to the 4th nearest neighbour of a point is $\varepsilon_i = 0.420$. In the X-marginal coordinate, there are 12 points with $|x_i - x_j| < 0.420$ and 1 point with $|x_i - x_j| = 0.420$. State the correct marginal count $N_{x,i}$ to be used in the KSG formula.
[Ans. $N_{x,i} = 12$ points]

8. Suppose a 5-seed benchmark yields rank differences between grok epochs and predictor event epochs of $d_s = (0, 1, 0, -1, 0)$. Calculate Spearman's rank correlation coefficient ρ and state whether it passes the benchmark threshold (≥ 0.90).
[Ans. $\rho = 0.90$; PASS]

9. For Seed 4, the observed grok epoch is G = 23,747 and the peak epoch is P = 32,500. Determine the timing lag and relative lead.
[Ans. Timing lag = +8,753 epochs; Relative lead = –0.3686]